HW 6

① $X(e^{j\omega}) = ?$ $\quad x[n] = n\alpha^n \mu[n+2]$, $|\alpha| < 1$

$$\hookrightarrow = n\,g[n] \longleftrightarrow j\,\frac{dG(e^{j\omega})}{d\omega}$$

$$\alpha^n \mu[n] \longleftrightarrow \frac{1}{1-\alpha e^{-j\omega}} \Rightarrow \alpha^{n+2}\mu[n+2] = \frac{e^{j2\omega}}{1-\alpha e^{-j\omega}}$$

$$\Rightarrow \alpha^2(\alpha^n \mu[n+2]) = \frac{e^{j2\omega}}{1-\alpha e^{-j\omega}} = \frac{\alpha^{-2}e^{j2\omega}}{1-\alpha e^{-j\omega}}$$

$$\hookrightarrow \frac{d\left(\frac{\alpha^{-2}e^{j2\omega}}{1-\alpha e^{-j\omega}}\right)}{d\omega} = \frac{(1-\alpha e^{-j\omega})(\alpha^{-2}(2j)e^{j2\omega}) - (\alpha^{-2}e^{j2\omega})(\alpha j e^{-j\omega})}{(1-\alpha e^{-j\omega})^2}$$

$$X(e^{j\omega}) = j\,\frac{2j\alpha^{-2}e^{2j\omega} - 2j\alpha^{-1}e^{j\omega} - j\alpha^{-1}e^{j\omega}}{(\quad)^2} = -\frac{2\alpha^{-2}e^{j2\omega} + 3\alpha^{-1}e^{j\omega}}{(1-\alpha e^{-j\omega})^2}$$

② $y[n] = \mu[n+2] - \mu[n-3]$ $\qquad \mu[n] = \frac{1}{1-e^{-j\omega}}$

$X(e^{j\omega}) = ?$ $\qquad \mu[n+2] = \frac{e^{j2\omega}}{1-e^{-j\omega}}$ $\qquad \mu[n-3] = \frac{e^{-j3\omega}}{1-e^{-j\omega}}$

$$X(e^{j\omega}) = \frac{e^{j2\omega}}{1-e^{-j\omega}} - \frac{e^{-j3\omega}}{1-e^{-j\omega}}$$

③ $x[n] = \{2, 3, -1, 0, -4, 3, 1, 2, 4\}$ $\quad -2 \le n \le 6$

a.) $X(e^{j0}) = \sum_{-2}^{6} x[n]\, e^{-j0n} = \underline{10}$

b.) $X(e^{j\pi}) = \sum_{-2}^{6} x[n]\, e^{-j\pi n} = 2e^{j2\pi} + 3e^{j\pi} - 1e^{0} + 0e^{-j\pi} - 4e^{-j2\pi} + 3e^{-j3\pi} + 1e^{-j4\pi}$
$+ 2e^{-j5\pi} + 4e^{-j6\pi}$

$= 2 - 3 - 1 + 0 - 4 - 3 + 1 - 2 + 4 = \underline{-6}$

c.) $\int_{-\pi}^{\pi} X(e^{j\omega})\, d\omega \Rightarrow x[n] = \frac{1}{2\pi}\int_{-\pi}^{\pi} X(e^{j\omega})\, e^{j\omega n}\, d\omega \Rightarrow 2\pi\, x[n] = \int_{-\pi}^{\pi} X(e^{j\omega}) e^{j\omega n}\, d\omega$

$\int_{-\pi}^{\pi} X(e^{j\omega})\, d\omega = 2\pi\, x[0] = \underline{-2\pi}$

d.) $\int_{-\pi}^{\pi} |X(e^{j\omega})|^2\, d\omega = \int_{-\pi}^{\pi} X(e^{j\omega})\, X^*(e^{j\omega})\, d\omega = 2\pi \sum_{-2}^{6} |x[n]|^2$

$= 2\pi\,(2^2 + 3^2 + 1^2 + 0^2 + 4^2 + 3^2 + 1^2 + 2^2 + 4^2) = \underline{120\pi}$

e.) $\int_{-\pi}^{\pi} \left| \frac{dX(e^{j\omega})}{dx} \right|^2 d\omega = 2\pi \sum_{-2}^{6} n^2 x^2[n]$

$= 2\pi\,(2^2 2^2 + 1^2 3^2 + 0^2 1^2 + 1^2 0^2 + 2^2 4^2 + 3^2 3^2 + 4^2 1^2 + 5^2 2^2 + 6^2 4^2)$

$= 2\pi\,(16 + 9 + 0 + 0 + 64 + 81 + 16 + 100 + 576)$

$= 2\pi\,(862) = \underline{1724\pi}$